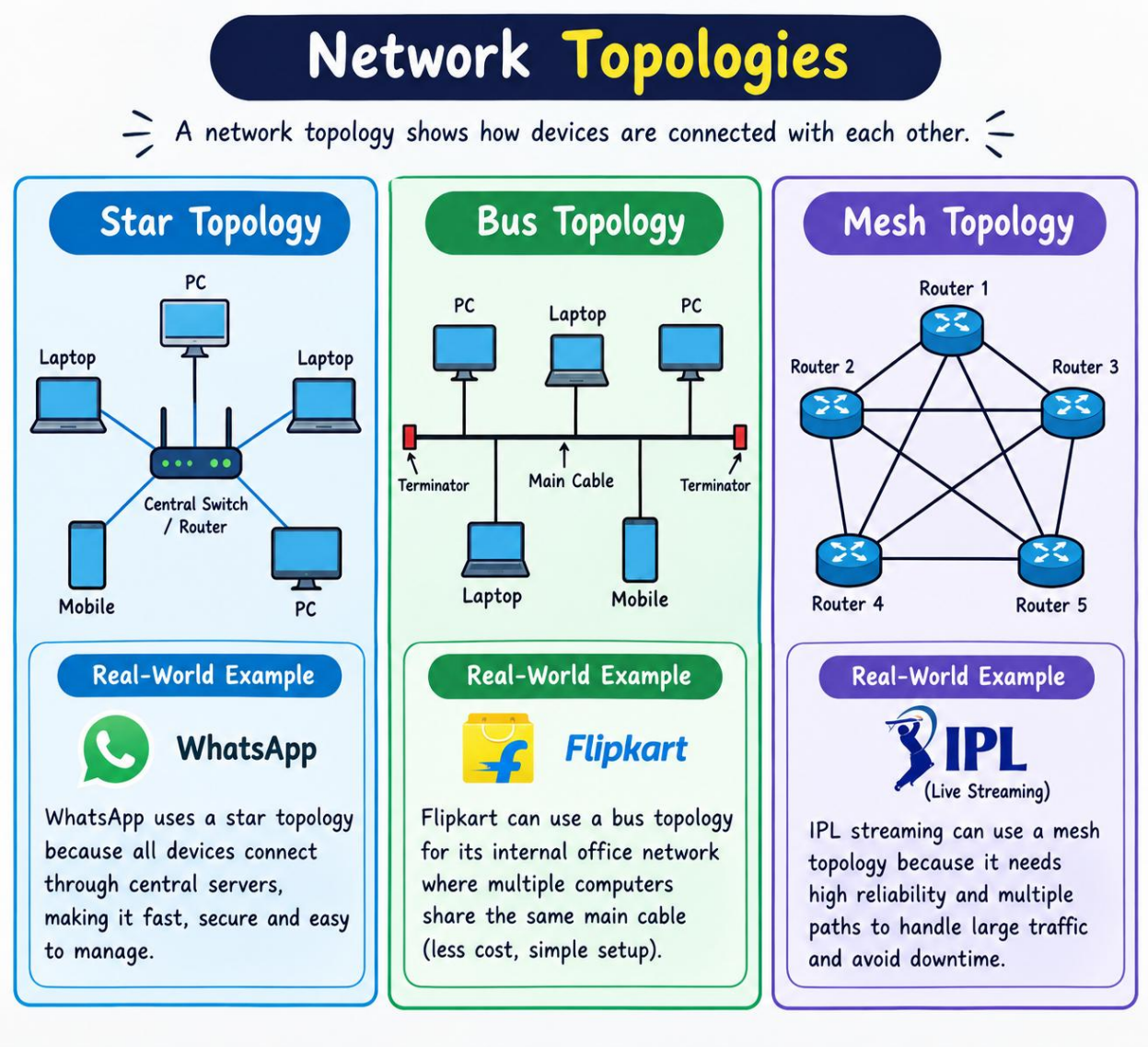
1. **Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.**



2. **Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).**

Topology	Pros (Advantages)	Cons (Disadvantages)	Best Use Case + App Example
1. Star Topology	1. Easy to install. 2. Easy to find problems. 3. One device failing does not affect others. 4. Easy to add new devices.	1. Needs more cables. 2. Costs more than bus topology. 3. If the central switch fails, the network stops. 4. Depends on the central device.	Best use: Schools, offices, and computer labs. Example: Instagram services can use a star-like setup in a local office network, where devices connect through a central switch.
2. Bus Topology	1. Uses less cable. 2. Low installation cost. 3. Simple for small networks. 4. Does not need a central switch.	1. If the main cable fails, the network may stop. 2. Difficult to find faults. 3. Performance becomes slow with heavy traffic. 4. Adding many devices can be difficult.	Best use: Small, simple, or older computer networks. Example: A small office network used for basic file sharing. Zomato's actual network design is not necessarily bus topology.
3. Ring Topology	1. Devices are connected in a circular path. 2. Data follows an organized route. 3. Can reduce data	1. A single link or device failure can disrupt a basic ring. 2. Adding devices can interrupt the network. 3. Troubleshooting	Best use: Specialized networks and some communication systems. Example: A ring-based office communication

	collisions in some designs. 4. Each device gets a chance to send data in token-based systems.	can be difficult. 4. Data may need to pass through several devices	network. Paytm's actual network is not necessarily ring topology.
4. Mesh Topology	1. Has multiple paths for data. 2. Very reliable. 3. One link failing may not stop communication. 4. Can handle failures better with alternative routes.	1. Expensive to install. 2. Needs more cables and connections. 3. Complex to manage. 4. Requires more setup and maintenance.	Best use: Networks needing high reliability. Example: Internet routing and some wireless mesh networks can provide alternative paths for data.

3. **Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.**

I would choose Mesh Topology for a multiplayer mobile game like PUBG or Free Fire.

1. Speed: Multiple network paths can help data travel quickly between servers.
2. Reliability: If one connection fails, another path may keep the game running.
3. Cost: It is expensive because it needs more connections and network equipment.
4. Overall: Mesh topology is a good choice when reliability is important, but the game also needs well-designed servers and fast internet to reduce lag.

4. **Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines. Consider factors like cost, ease of troubleshooting, and future expansion.**

